Skills Summary

Linear Equations with Distribution and Fractions

Use this reference while completing your worksheet or when studying.

The order that always works: **(1)** clear fractions by multiplying *every* term on *both* sides by the LCD, **(2)** distribute across any brackets, **(3)** collect variable terms on one side and numbers on the other, **(4)** divide to isolate x — then substitute back into the original equation to check.

1. Distribute first, then undo (distribute-then-solve)

Distributive property: $a(b+c) = ab + ac$, and $a(b-c) = ab - ac$.

The factor outside multiplies **every** term inside — including the second one, with its sign. A negative factor changes both signs.

Example: Solve $4(x-3) = 20$

Step 1. The variable is trapped inside brackets, so distribute the 4 over both terms before undoing anything.

Step 2. $4x - 12 = 20$, so $4x = 32$ and $x = 32 \div 4$. $\Rightarrow$ $\boldsymbol{x = 8}$ Check: $4(8-3) = 20$ ✓

Try it: Solve $5(x+2) = 45$

check: $\boldsymbol{x = 7}$

2. Clear the fractions with the LCD (clear-fractions)

Rule: Find the LCD of every denominator, then multiply **every term on both sides** by it — whole-number terms included.

$\dfrac{x}{a}$ multiplied by the LCD L becomes $\dfrac{L}{a}x$. If a fraction bar covers a whole numerator, keep that numerator in brackets: $L \cdot \dfrac{x+3}{2} = \dfrac{L}{2}(x+3)$.

Example: Solve $\dfrac{x}{2} + \dfrac{x}{6} = 4$

Step 1. Two different denominators make this hard to collect, so multiply everything by the LCD of 2 and 6, which is 6.

Step 2. $3x + x = 24$, so $4x = 24$. $\Rightarrow$ $\boldsymbol{x = 6}$ Check: $3 + 1 = 4$ ✓

Try it: Solve $\dfrac{x}{3} + \dfrac{x}{4} = 7$

check: $\boldsymbol{x = 12}$

3. Variables on both sides (variables-both-sides)

Rule: Distribute on each side first, then move the *smaller* variable term across so the coefficient you are left with is positive.

Whatever you subtract from one side, subtract from the other — that is what keeps the equation balanced.

Example: Solve $2(x+5) = 4x - 6$
Step 1. Both sides carry an x, so distribute first and then gather the variable on the side where it stays positive.
Step 2. $2x + 10 = 4x - 6$; subtract $2x$: $10 = 2x - 6$; add 6: $16 = 2x$. $\Rightarrow$ $\boldsymbol{x = 8}$ Check: $2(13) = 26$ and $4(8) - 6 = 26$ ✓

Try it: Solve $3(x-1) = x + 9$

check: $\boldsymbol{x = 6}$

4. Rewriting into an equivalent form (rewrite-equivalent)

Rule: Multiplying an expression by the LCD produces an **equivalent** expression only if every term is multiplied.
$L\left(\frac{x}{a} + \frac{b}{c}\right) = \frac{L}{a}x + \frac{Lb}{c}$. Doing this to one side of an equation means doing it to the other side too.

Example: Multiply $\frac{x}{3} - \frac{1}{2}$ by the LCD 6
Step 1. The denominators are 3 and 2; their LCD is 6, and multiplying by it clears both at once — no solving is involved here, only rewriting.
Step 2. $6 \cdot \frac{x}{3} = 2x$ and $6 \cdot \frac{1}{2} = 3$. $\Rightarrow$ $\boldsymbol{2x - 3}$

Try it: Multiply $\frac{x}{5} + \frac{3}{4}$ by the LCD 20

check: $\boldsymbol{4x + 15}$

(!) Watch out: A minus sign in front of brackets or in front of a fraction bar reaches *every* term behind it: $-(x-5) = -x + 5$. And when you clear fractions, the whole-number terms get multiplied too — forgetting them is the single most common lost mark.